

Homework 3

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Theorem 1. *Using the concept of open covers (and explicitly avoiding the Bolzano–Weierstrass Theorem), prove that every bounded infinite set has a limit point.*

Proof. Let $A \subset \mathbb{R}$ be bounded and infinite. So there exists $M > 0$ such that

$$A \subseteq [-M, M].$$

Note that $[-M, M]$ is compact in $\mathbb{R}$ by the Heine-Borel Theorem as it is closed and bounded.

We aim for a contradiction by assuming that A has no limit point. By definition, if A has no limit point, then for every $x \in A$, there exists some $\varepsilon_x > 0$ such that the open interval

$$(x - \varepsilon_x, x + \varepsilon_x) \cap A = \{x\}.$$

That is, each point $x \in A$ lies in some open interval that contains *no other* points of A .

Now, construct an open cover of the compact set $[-M, M]$ as follows:

$$\mathcal{U} = \{U_x : x \in A\} \cup \{\mathbb{R} \setminus A\},$$

where each

$$U_x = \left(x - \frac{\varepsilon_x}{2}, x + \frac{\varepsilon_x}{2}\right).$$

Observe that:

- For each $x \in A$, the interval U_x is open and contains x .
- By construction, $U_x \cap A = \{x\}$ (we shrunk the interval from ε_x to $\varepsilon_x/2$ so it still contains no other points of A).
- The set $\mathbb{R} \setminus A$ is also open.

This family $\mathcal{U}$ covers $[-M, M]$ because:

$$\left(\bigcup_{x \in A} U_x\right) \cup (\mathbb{R} \setminus A) = \mathbb{R}.$$

In particular, $[-M, M]$ is certainly covered by $\mathcal{U}$.

Since $[-M, M]$ is compact, every open cover of $[-M, M]$ must admit a *finite subcover*. Therefore, there exist finitely many points $x_1, x_2, \dots, x_n \in A$ such that

$$\{U_{x_1}, U_{x_2}, \dots, U_{x_n}, \mathbb{R} \setminus A\}$$

is still a cover for $[-M, M]$.

As $A \subseteq [-M, M]$, this finite subcover also covers A . Because of the construction of the finite subcover, every open set in the finite subcover either covers one point in A , or no point in A at all. Thus, there are finite points in A . But, A was infinite, and thus we have a contradiction.

Therefore, our assumption that A has no limit point is false. It follows that A *must have a limit point*. $\square$

References

I came up with this proof and all its parts by myself. Then, I used ChatGPT-o1 to convert it into a LaTeX document for readability.